

American Economic Association

Output Growth, the Real Wage, and Employment Fluctuations

Author(s): Zvi Hercowitz and Michael Sampson

Source: *The American Economic Review*, Vol. 81, No. 5 (Dec., 1991), pp. 1215-1237

Published by: [American Economic Association](#)

Stable URL: <http://www.jstor.org/stable/2006914>

Accessed: 05/10/2014 10:32

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at
<http://www.jstor.org/page/info/about/policies/terms.jsp>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.



American Economic Association is collaborating with JSTOR to digitize, preserve and extend access to *The American Economic Review*.

<http://www.jstor.org>

Output Growth, the Real Wage, and Employment Fluctuations

By ZVI HERCOWITZ AND MICHAEL SAMPSON*

This paper is an attempt to contribute to the integration of business-cycle analysis with long-term growth. A real-business-cycle model with endogenous growth is developed and estimated with U.S. data. In the present framework, wage movements do not have to be transitory to generate fluctuations in labor effort. The reduced form is a constrained bivariate output/hours (or real-wage/hours) VAR process. The bivariate setup provides a useful framework for analyzing the persistence of output fluctuations, given that the theory implies that hours of work contain information about future output movements. (JEL E32)

The analysis of business cycles requires that the researcher come to grips with the presence of long-term growth in macroeconomic time-series. The traditional approach has been to assume that growth occurs as the result of exogenous technological progress, which can be captured by a deterministic function of time. A measurement of the business cycle is then the residuals obtained from fitting a deterministic trend to the series.

An alternative approach is to treat long-term growth as endogenous, as in Robert King and Sergio Rebelo (1986) and King et al. (1988a, b). These models follow Robert Lucas (1988) and Paul Romber (1986) in the incorporation of mechanisms generating sustained growth without exogenous technological change. A major implication of this framework is that temporary shocks have permanent effects. Hence, the decomposition of a series into a deterministic trend and a stationary cycle is inconsistent with this theory.

The present paper pursues this line of research, which integrates the analysis of business fluctuations with long-term growth considerations. A real-business-cycle model with endogenous growth is developed. It predicts that both output and the real wage (in logarithms) are generated by processes with unit roots, while hours of work follow a stationary process. In the present model, fluctuations in hours of work reflect the changing relative labor productivity in home and market activities. A consequence of this setup is that real wage fluctuations do not have to be temporary in order to generate fluctuations in labor effort. A related implication is that the secular increase in the real wage is consistent with stationary hours of work, even though there is no income effect involved.

The empirical analysis focuses on two alternative bivariate systems derived from the model: output/hours, and the real wage/hours. The model predicts that both systems follow constrained bivariate vector autoregressions. These processes are estimated with post-World War II U.S. data, and the restrictions are tested. Structural parameters related to labor supply and capital accumulation, as well as the parameters of the joint process of two exogenous shocks affecting this economy, are estimated. One disturbance is a standard shock to the production function, as is usually found in the real-business-cycle models, and the other is a disturbance to capital accumulation.

*Tel-Aviv University, Tel-Aviv 69978, Israel, and Concordia University, 1455 Maisonneuve Blvd., W. Montreal, Quebec H3G 1M8, Canada, respectively. We thank Jeremy Greenwood and the referees for helpful discussions and comments. The first author acknowledges financial support from the David Horowitz Institute and the Foerder Institute for Economic Research at the Tel-Aviv University.

The bivariate output/hours setup provides a useful framework for analyzing the persistence of output fluctuations, given that the theory implies that hours contain information about future output growth. In this sense, the present paper extends the work of Charles Nelson and Plosser (1982), John Campbell and N. Gregory Mankiw (1987), Mark Watson (1986), John Cochrane (1988), and others, who used univariate frameworks. The question of the permanence of output fluctuations can also be posed in terms of how the movements in actual output relate to those in its long-run stochastic trend. The procedure adopted here to obtain a stochastic trend is a multivariate version of the method suggested by Stephen Beveridge and Nelson (1981), as discussed in King et al. (1989).

The paper proceeds as follows. In Section I, the setup of the model is described, and the special features of tastes and technology are discussed. The solution to the model is presented in Section II, and its implications for the comovement of real wages and labor effort are discussed in Section III. The estimation is reported in Section IV, and Section V addresses the calculation of the stochastic trend in output and the persistence of fluctuations. Section VI contains concluding remarks.

I. The Setup of the Model

The framework is a stochastic-growth model of the type used by Finn Kydland and Edward Prescott (1982) and John Long and Plosser (1983). As in the latter paper, the present model has a log-linear structure, so that a closed-form solution can be obtained. Here, however, hours of work are not constant in equilibrium, and capital does not fully depreciate in one production period.

The economy is composed of a large and constant number of identical households and identical firms interacting in a competitive environment. The technology and preferences are specified and discussed next. Special attention is given to the aspects in which the present specification diverges from that usually used in real-business-cycle models.

The representative firm produces output according to the technology

$$(1) \quad Y_t = F(K_t, H_t, L_t, z_{1t}) \\ = A_0 K_t^\alpha (H_t L_t)^{1-\alpha} \exp(z_{1t}) \\ 0 < \alpha < 1$$

where z_{1t} is a productivity shock, K_t is physical capital in productivity units, H_t is an index of knowledge, and L_t is labor input in time units. Hence, the accumulation of human capital has the effect of Harrod-neutral technological progress. The shock z_{1t} follows a stationary process to be specified below.

The capital stock evolves as

$$K_{t+1} = K_t G(I_t/K_t) \exp(z_{2t+1}) \\ G' > 0 \quad G'' < 0$$

where I_t is the amount of resources devoted both to investment that increases the quantity of capital and to research that improves its quality. This is the type of capital-evolution equation used by Lucas and Prescott (1971). Unlike the standard linear form (i.e., $K_{t+1} = K_t[1-d] + I_t$), it exhibits decreasing returns, which can be interpreted as reflecting adjustment costs in increasing the volume of capital or diminishing returns in research activities. Capital accumulation is subject to the stationary disturbance z_{2t} . One may think of the oil shocks of the 1970's as being captured by this disturbance, as drastic increases in energy prices rendered part of the capital stock economically obsolete.

The specific form adopted for the function G is $(I_t/K_t)^{1-\delta}$, $0 < \delta < 1$, so that the capital-evolution equation becomes¹

$$(2) \quad K_{t+1} = A_1 K_t (I_t/K_t)^{1-\delta} \exp(z_{2t+1}).$$

¹A drawback of (2) is that if I_t is zero, K_{t+1} is zero as well. In any event, the present model will always generate positive investment.

An alternative interpretation can be given to (2) by rewriting it as

$$K_{t+1} = A_1 K_t^\delta I_t^{1-\delta} \exp(z_{2t+1}).$$

The parameter δ can be associated then with the relative quality of old capital relative to new investment goods.

The shocks z_{1t} and z_{2t} are assumed to follow the vector autoregressive process $\phi(B)\mathbf{z}_t = \mathbf{a}_t$ where $\mathbf{z}_t = (z_{1t}, z_{2t})'$, $\mathbf{a}_t$ is the vector white noise $(a_{1t}, a_{2t})'$ and $\phi(B)$ is a 2×2 matrix polynomial of order p in the backshift operator B .

As in Kenneth Arrow (1962) and Romer (1986), knowledge is assumed to grow proportionally to and as a by-product of the accumulated investment and research activities in the economy:

$$(3) \quad H_t = \bar{K}_t$$

where $\bar{K}_t$ is the average capital stock across firms.² Thus, the production function of the representative firm—equation (1)—displays increasing returns at the social level, but the model is consistent with competitive equilibrium since each firm takes H_t as given.

The representative household maximizes

$$E_t \sum_{j=0}^{\infty} \beta^j U[C_{t+j}, H_{t+j}(L - J(L_{t+j}))] \quad 0 < \beta < 1$$

where C_t is the flow of consumption goods per household, L is the flow endowment of time, and $H_t(L - J(L_t))$ measures effective leisure time as input in home activities, which in turn produce utility. The function $J(\cdot)$, with $J'(\cdot) > 0$, $J''(\cdot) > 0$, represents a fatigue effect that reduces effective leisure time. Following Gilbert Ghez and Gary Becker (1975) and James Heckman (1976), we incorporate the notion that knowledge increases the productivity of the input of time in home activities. Hence, H_t not only

increases market labor productivity, but also the marginal disutility from labor supply. This effect is captured by

$$(4) \quad U = \ln[C_t - H_t L_t^{1+\omega}] \quad \omega > 0.$$

This form implies that the direct positive effect of H_t (which is not a choice variable) on utility is neglected. The corresponding formulation including this effect would have been

$$(5) \quad U = \ln[C_t + H_t(L - L_t^{1+\omega})].$$

The form in (4) was chosen instead of (5) because it allows for a closed-form solution to the model. Given the important role that this type of criterion function plays in the paper, it is appropriate to elaborate first on the economic implications of (4) and (5) and then to assess how the approximation taken in (4) is likely to affect the results.

An important property of both (4) and (5) is that the marginal rate of substitution between consumption and labor supply is independent of the consumption level. Hence, L_t can be solved independently of the intertemporal optimization over consumption and saving. This, however, would hold not only for the logarithmic function, but also for any function applied to the composite $C_t - H_t L_t^{1+\omega}$ or to $C_t + H_t(L - L_t^{1+\omega})$. Defining W_t as the real wage at time t , the condition for optimal labor supply for both (4) and (5) is

$$(6) \quad -\frac{U_\ell(t)}{U_c(t)} = (1 + \omega) H_t L_t^\omega = W_t.$$

Labor supply is a stationary variable if H_t and W_t have the same stochastic trend. This holds in the model. The accumulation of human capital affects not only the real wage, but also the opportunity cost of supplying labor. The important characteristic of (4) and (5) is that there is no income effect at work. Labor supply remains stationary as the real wage increases over time, because of parallel increasing productivity in home activities.

²The assumption of proportionality in equation (3) is crucial for generating balanced growth in this model.

The criterion functions (4) and (5) also have important implications regarding the effect of the real wage on labor effort. A standard utility function, like the Cobb-Douglas function, has the important characteristic that only *temporary* movements in the real wage affect labor supply. Permanent movements do not alter labor effort because they generate offsetting substitution and income effects. It can be observed in (6) that, under the present utility function, real wage movements have a positive effect on labor supply regardless of whether they are permanent or transitory. What matters for this response is whether the increased labor productivity driving the real wage affects home productivity (H_t) contemporaneously or with a lag. Labor supplied in the market is, in the present setup, a result of shifting relative productivities at work and at home.³

We turn now to the implications of using the form in (4) rather than that in (5). As mentioned above, (4) neglects the direct effect of the externally evolving human capital on utility, reflected in $H_t L$. As seen above, since this term does not affect the marginal rate of substitution between consumption and leisure, the conditions for optimal labor supply following from (4) and (5) are identical. Another marginal rate of substitution is that between current and future consumption, which determines the investment decision. The term $H_t L$, which is not a choice variable, reduces the marginal utility from current consumption, but at the same time, it also reduces that of future consumption. Hence, the exclusion of $H_t L$ does not systematically bias the investment decision. As a consequence, in a deterministic version of the model, the balanced-growth path is not affected at all by the exclusion of $H_t L$. This is shown in Appendix A. The question now is how the investment decision following from (4) approximates the investment decision from (5) in the stochastic framework. Appendix A

³See Jeremy Greenwood and Hercowitz (1990) for an analysis of home and market productivity of labor with home capital accumulation.

describes simulations of the exact choice. Our conclusion from those exercises is that the adoption of (4) instead of (5) should have little impact on the investment decision and the rest of the model.

Finally, the choices in the economy are constrained by

$$(7) \quad Y_t = C_t + I_t.$$

II. Solution of the Model

The interaction of optimizing households and firms in competitive equilibrium is analyzed by solving the following representative agent's dynamic problem:

$$\begin{aligned} V(K_t, \bar{K}_t, \underline{z}_t) \\ = \max_{\{C_t, I_t, L_t\}} \left\{ \ln(C_t - \bar{K}_t L_t^{1+\omega}) \right. \\ \left. + \beta E_t V(K_{t+1}, \bar{K}_{t+1}, \underline{z}_{t+1}) \right\} \end{aligned}$$

where $\underline{z}_t = (z_t, z_{t-1}, \dots, z_{t-p+1})$, subject to:

$$C_t = Y_t - I_t$$

$$Y_t = A_0 K_t^\alpha (\bar{K}_t)^{1-\alpha} L_t^{1-\alpha} \exp(z_{1t})$$

$$K_{t+1} = A_1 K_t^\delta I_t^{1-\delta} \exp(z_{2t+1}).$$

Additionally, the individual agent takes the evolution of the average level of capital in the economy as given by the process

$$\begin{aligned} (2') \quad \bar{K}_{t+1} &= A_2 (\bar{K}_t)^{\tau_1} \exp \left[\sum_{j=0}^p \tau_{2j} z_{1,t+1-j} + \tau_{3j} z_{2,t+1-j} \right] \\ \tau_1 &< 1/\beta. \end{aligned}$$

Since this process is taken as given, the individual agent does not recognize the effect of his own decisions on $\bar{K}_{t+1}$.

For any arbitrary set of coefficients in (2') (satisfying the condition $\tau_1 < 1/\beta$ for finiteness of utility), the model can be solved by conjecturing that the value function is of

the form

$$V(K_t, \bar{K}_t, \mathbf{z}_t) = D_0 + D_1 \ln K_t + D_2 \ln \bar{K}_t + \sum_{j=0}^{p-1} \mathbf{D}_j^* \mathbf{z}_{t-j}$$

where $\mathbf{D}_j^*$ is a 1×2 vector.

The solution is:

$$(8) \quad L_t = A_\ell K_t^{\frac{\alpha}{\alpha+\omega}} (\bar{K}_t)^{\frac{-\alpha}{\alpha+\omega}} \times \exp(z_{1t})^{\frac{1}{\alpha+\omega}}$$

where

$$(9) \quad A_\ell = \left[A_0 \left(\frac{1-\alpha}{1+\omega} \right) \right]^{\frac{1}{\alpha+\omega}} A_y K_t^{\frac{\alpha(1+\omega)}{\alpha+\omega}} (\bar{K}_t)^{\frac{\omega(1-\alpha)}{\alpha+\omega}} \times \exp(z_{1t})^{\frac{1+\omega}{\alpha+\omega}}$$

where

$$(10) \quad A_y = \left[A_0^{1+\omega} \left(\frac{1-\alpha}{1+\omega} \right)^{1-\alpha} \right]^{\frac{1}{\alpha+\omega}} I_t = b Y_t$$

where

$$(11) \quad b = \frac{\beta D_1 (1-\delta)}{1 + \beta D_1 (1-\delta)} \left(\frac{\alpha + \omega}{1 + \omega} \right) C_t = (1-b) Y_t$$

$$(12) \quad K_{t+1} = A_k K_t^{\delta + \frac{\alpha(1+\omega)}{\alpha+\omega}(1-\delta)} \times (\bar{K}_t)^{\frac{\omega(1-\alpha)(1-\delta)}{\alpha+\omega}} \times \exp(z_{1t})^{\frac{(1+\omega)(1-\delta)}{\alpha+\omega}} \times \exp(z_{2t+1})$$

where

$$A_k = A_1 (b A_y)^{1-\delta}.$$

The coefficients D_1 and D_2 are

$$D_1 = \frac{\alpha(1+\omega)/(\alpha+\omega)}{1 - \beta \left(\frac{\alpha + \omega [\delta + \alpha(1-\delta)]}{\alpha + \omega} \right)} > 0$$

$$D_2 = \frac{\omega(1-\alpha)[1 + \beta D_1(1-\delta)]/(\alpha + \omega)}{1 - \beta \tau_1} > 0.$$

Since D_1 is positive (and $0 < \alpha < 1$), this implies that $0 < b < 1$. The other coefficients of the value function, which also can be calculated, are not relevant for our purposes. Substituting D_1 into equation (9) yields⁴

$$(10') \quad I_t = \frac{\alpha \beta (1-\delta)}{1 - \delta \beta} Y_t.$$

As in the similar contexts analyzed by Romer (1986), King and Rebelo (1986), and King et al. (1988a, b), the model generates an endogenous "engine of growth." This role is taken in the present model by the capital-stock process in equation (12). Consider first the case in which the externality does not exist and hence $\bar{K}_t$ drops from equation (12). Since the exponent of K_t is less than 1, this would imply that the stochastic difference equation in (12) is a stationary process. Therefore, no long-term growth would occur in this economy.

The presence of the externality implies that K_{t+1} depends not only on K_t but also on $\bar{K}_t$. Now, since all firms are identical the

⁴Appendix A reports numerical simulations of the model using the exact utility function (5), under two extreme assumptions about z_{1t} : white noise and a random walk. The solutions for L_t and Y_t are identical to those in (8) and (9). The investment choice is not proportional to output, as in (10'). However, the deviations from proportionality seem reasonably small.

equality $\bar{K}_t = K_t$ holds. Then, equation (12) becomes

$$(12') \quad K_{t+1} = A_k K_t \exp(z_{1t})^{\frac{(1+\omega)(1-\delta)}{\alpha+\omega}} \times \exp(z_{2t+1}).$$

Note in (12') that the capital stock (in logs) contains an autoregressive unit root. Hence, the stationary disturbances z_{1t} and z_{2t+1} have permanent effects on the capital stock. As pointed out by Bennett McCallum (1989) a condition for the existence of unit roots in endogenous-growth models is that constant returns hold exactly. It can be observed in (12) that the combined coefficient of K_t and $\bar{K}_t$ being 1 depends on the constant-returns assumption (i.e., that both pairs of exponents (α and $1-\alpha$ from the production function; δ and $1-\delta$ from the capital-formation equation) add up to 1. If constant returns were only an approximation, this model would imply two things: (i) the autoregressive coefficient in capital would be close to but not equal to 1, and (ii) the growth rates of capital would either continuously increase over time, or reach an eventual average of zero. In other words, exact constant returns are a necessary condition for both unit roots and balanced growth.

Since $\bar{K}_t = K_t$, the process for the average capital stock in (2') coincides with the process for K_t in (12'). Hence, the arbitrary parameters of (2') should be identical to the corresponding coefficients in (12'). In particular $\tau_1 = 1$, satisfying the condition $\tau_1 < 1/\beta$.⁵

We turn now to consider the solution to the two variables of main interest: output and hours of work. Since output depends on the capital stock, the nonstationarity of the latter is absorbed by output as well. With

⁵Comparing (2') and (10'), it follows that $\tau_{2,1} = (1+\omega)(1-\delta)/(\alpha+\omega)$, $\tau_{3,0} = 1$, $A_2 = A_k$, and the rest of the coefficients in (2') are equal to zero.

$\bar{K}_t = K_t$, equation (9) becomes

$$(9') \quad Y_t = A_y K_t \exp(z_{1t})^{\frac{1+\omega}{\alpha+\omega}}$$

where the "trend" in output is generated by the accumulation of K_t .

When $K_t = \bar{K}_t$ the process for hours of employment in equation (8) becomes

$$(8') \quad L_t = A_\ell \exp(z_{1t})^{\frac{1}{\alpha+\omega}}.$$

The capital stock drops from the labor equation because of the two opposite forces that it exerts on labor supply. Since knowledge affects proportionally both the market and the nonmarket marginal productivities of labor, productivity at home rises as much as in the marketplace.

To establish the link between output and employment fluctuations in an empirically useful form, the solution for output is expressed in terms of its rate of change. From equations (8'), (9'), and (12') and using lowercase letters for natural logs, it follows that

$$(13) \quad \Delta y_t = \mu + \left(\frac{1+\omega}{\alpha+\omega} \right) (1-\delta B) z_{1t} + z_{2t} \quad \mu = \ln A_k$$

$$(14) \quad l_t = \ln(A_\ell) + \frac{1}{\alpha+\omega} z_{1t}.$$

These two equations form the main structural empirical model to be estimated.

Equation (13) shows that the disturbance to capital accumulation, z_{2t} , has a permanent effect on output, while the effect of the production-function disturbance, z_{1t} , is partially reversed next period according to the parameter δ ($0 < \delta < 1$). Note that the theory rules out the possibility of the existence of a deterministic trend in output. For this to be the case, the variance of z_{2t} would have to be zero, and more importantly, the moving-average coefficient δ would have to

be 1, implying that investment does not contribute at all to future productive capacity.

Finally, it is interesting to consider the implications of the adjustment costs embodied in the capital-evolution equation (2) for the behavior of Tobin's q in the present framework.

From a formal exercise of pricing old capital in this model it can be shown that the optimizing agents equate the relative price of old capital in terms of new capital (Tobin's q) to the ratio of the marginal contributions of K_t and I_t to K_{t+1} . Specifically:

$$q = \frac{\partial K_{t+1} / \partial K_t}{\partial K_{t+1} / \partial I_t} = \left(\frac{\delta}{1 - \delta} \right) \frac{I_t}{K_t}.$$

As shown by Fumio Hayashi (1982) in the general context of constant returns, Tobin's q is here a positive function of the investment:capital ratio. In the standard linear case, K_{t+1} is the sum of end-of-period capital and new investment, and hence if there are no adjustment costs q should always be equal to 1.

From (9') and (10) it follows that

$$q = [\delta / (1 - \delta)] I_t / K_t \\ = [\delta / (1 - \delta)] b A_y \exp \left(\frac{1 + \omega}{\alpha + \omega} z_{1t} \right).$$

Tobin's q is, as usual, stationary, but its mean depends on the parameters of the model that determine the average I/K ratio. In particular, it is a feature of the present framework, which combines an endogenous growth mechanism and adjustment costs, that q depends on the constant in the production function. The constant A_0 , representing the general efficiency of production in the economy, enters A_y [equation (9)] and affects positively both the average growth rate of the economy μ [equations (9), (12), and (13)], and the I/K ratio. Given that A_0 is not identifiable, this model does not place a restriction on the average value of q .

III. The Real Wage and Labor Effort

This model generates fluctuations in labor effort through a different channel than the intertemporal-substitution mechanism stressed originally by Lucas and Leonard Rapping (1969) and incorporated in the real-business-cycle models. The intertemporal-substitution mechanism has to do with the response of labor supply to the deviations of the real wage from the normal future level. Hence, for example, if the real wage follows a random walk this channel does not operate, and labor effort does not respond to real wage movements. This is not the case in the present model. To illustrate this, the solution of the model can be used to derive the real wage that would support the process for L_t derived from the model in a competitive environment.

Equalizing the real wage, W_t , to the marginal productivity of labor at the solution values yields

$$(15) \quad W_t = A_w K_t \exp(z_{1t})^{\frac{\omega}{\alpha + \omega}}$$

where

$$A_w = (1 - \alpha) A_0 A_\ell^{-\alpha}.$$

From the process of K_t in (12'), the first difference of the log of the real wage can be expressed as

$$(16) \quad \Delta w_t = \mu + \frac{1}{\alpha + \omega} \{ \omega + [1 - \delta(1 + \omega)] B \} z_{1t} + z_{2t}$$

where

$$\mu = \ln A_k.$$

To highlight the implications of the present specification, assume for a moment that $\delta(1 + \omega) = 1$, which is consistent with the theory, and that z_{1t} and z_{2t} are white noise. In this case, the real wage follows a random walk with drift. However, labor effort still comoves with the real wage according to equations (14) and (15).

The different implications of the present mechanism, relative to the intertemporal-substitution mechanism, is that what matters for labor supply here is the contemporaneous productivity differential in market and home activities, represented by the shock z_{1t} . As market productivity and, hence, the real wage change with a given realization of z_{1t} , labor supply reacts positively. Over time, the transitory shock generates the accumulation of capital and knowledge, which cancel each other as to the labor-supply decision. The increment in the real wage typically becomes permanent, given the accumulation of capital, but as the shock z_{1t} dissipates so does the labor fluctuation.

Another aspect of the relationship between real wages and hours worked is addressed by Lawrence Christiano and Martin Eichenbaum (1988). They stress the contrast between the positive correlation between real wages and hours, predicted by the existing real-business-cycle models, and the observation that the actual correlation is very weak. Christiano and Eichenbaum argue that it is important to incorporate shocks shifting labor supply, in addition to the usual productivity shocks affecting labor demand, for a model to conform with the observed correlations.

Given the present mechanism, by which human-capital accumulation can be seen as shifting labor supply, it is interesting to address the model's implications regarding the comovement of real wages and hours of work. For convenience, the relevant expressions are rewritten in first-difference form:

$$(16) \quad \Delta w_t = \mu + \frac{\omega}{\alpha + \omega} z_{1t} + \left[\frac{1 - \delta(1 + \omega)}{\alpha + \omega} \right] z_{1t-1} + z_{2t}$$

$$(14') \quad \Delta l_t = \frac{1}{\alpha + \omega} z_{1t} - \frac{1}{\alpha + \omega} z_{1t-1}.$$

For the case in which $\delta(1 + \omega) < 1$ holds, the coefficients of z_{1t-1} in the two equations have opposite signs. This implies that,

at times, the real wage and hours may be moving in opposite directions. This tends to reduce the positive correlation generated by the current z_{1t} .

What happens can be interpreted as movements of labor demand and supply schedules. Consider a positive productivity shock z_{1t} . Contemporaneously, labor demand increases, thus generating a positive comovement of hours and the real wage. Abstracting from subsequent disturbances, as the shock dissipates, the demand curve does not return to the initial position, because of the accumulation of new capital. This accumulation is larger the smaller is δ . Additionally, the parallel increase in H_t shifts the supply schedule to the left. The smaller the contraction in labor demand and the larger the contraction in supply, the more likely it is that the comovement of hours and real wages becomes negative as the shock dissipates.

IV. Estimation

This section presents the derivation of reduced-form representations for the model and the testable restrictions it imposes. Then, the empirical results are reported.

The structural form of the model, in (13) and (14), can be written as

$$(17) \quad \mathbf{x}_t = \mathbf{\Omega}(B) \mathbf{z}_t,$$

where

$$\mathbf{x}_t = [\Delta y_t - \mu, l_t - \ln A_t]'$$

$$\mathbf{\Omega}(B) = \begin{bmatrix} (1 + \omega)(1 - \delta B) & 1 \\ 1 & 0 \end{bmatrix}$$

and the factor $1/(\alpha + \omega)$ has been absorbed into z_{1t} (since α turns out to be not identifiable).

As mentioned in Section I, $\mathbf{z}_t$ is assumed to follow a vector autoregressive process of order p given by $\phi(B)\mathbf{z}_t = \mathbf{a}_t$, where $\mathbf{a}_t$ is independently and identically distributed (henceforth i.i.d.) $\mathcal{N}[0, \Sigma]$, $\Sigma = [\sigma_{ij}]$ for $i, j = 1, 2$, and $\phi(B)$ is a 2×2 matrix polynomial of order p with $\phi(0) = \mathbf{I}$.

From (17) it appears that $\mathbf{x}_t$ follows a vector ARMA($p, 1$) process if $\mathbf{z}_t$ is a VAR(p) process. However, $\mathbf{x}_t$ has a VAR($p + 1$) representation as well. This is so because $\det[\Omega(B)] = -1$, so that $\Omega(B)^{-1}$ is also a first-order polynomial (rather than being of the usual infinite order) given by

$$\Omega(B)^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & -(1 + \omega)(1 - \delta B) \end{bmatrix}$$

so that

$$(18) \quad \phi(B)\Omega(B)^{-1}\mathbf{x}_t = \mathbf{a}_t$$

and hence $\mathbf{x}_t$ follows a VAR($p + 1$) process. This simplifies the estimation work considerably.

However, (18) is not the usual VAR representation, since $\Omega(0)^{-1} \neq \mathbf{I}$. The standard representation, with the autoregressive operator normalized to be the identity matrix when evaluated at zero, is given by

$$\begin{aligned} \Omega(0)\phi(B)\Omega(0)^{-1}\Omega(0)\Omega(B)^{-1}\mathbf{x}_t \\ = \Omega(0)\mathbf{a}_t \end{aligned}$$

or

$$(19) \quad \tilde{\phi}(B)\tilde{\Omega}(B)^{-1}\mathbf{x}_t = \tilde{\mathbf{a}}_t$$

where

$$(20) \quad \tilde{\phi}(B) = \Omega(0)\phi(B)\Omega(0)^{-1}$$

$$\tilde{\Omega}(B) = \Omega(B)\Omega(0)^{-1}$$

$$\tilde{\mathbf{a}}_t \sim \text{i.i.d. } \mathcal{N}[0, \tilde{\Sigma}]$$

$$\tilde{\Sigma} = \Omega(0)\Sigma\Omega(0).$$

The polynomial $\tilde{\phi}(B)$ is also of order p with $\tilde{\phi}(0) = \mathbf{I}$; $\tilde{\Omega}(B)^{-1}$ is of order 1 with $\tilde{\Omega}(0)^{-1} = \mathbf{I}$ and is given by

$$(21) \quad \tilde{\Omega}(B)^{-1} = \begin{bmatrix} 1 & \lambda B \\ 0 & 1 \end{bmatrix}$$

$$\lambda = \delta(1 + \omega).$$

The representation in (19) can be viewed as the reduced form with reduced-form parameters: λ , $\tilde{\Sigma}$, and the $4 \times p$ coefficients in $\tilde{\phi}(B)$. The VAR($p + 1$) process is constrained by the model to be of the form $\tilde{\phi}(B)\tilde{\Omega}(B)^{-1}$ so that three testable restrictions are imposed on the VAR($p + 1$) operator. That is, if one estimates an unrestricted VAR($p + 1$) process, there would be $4(p + 1)$ free parameters, while the restricted VAR($p + 1$) depends only on $4p + 1$ free parameters: λ and $\tilde{\phi}(B)$.⁶

Unfortunately, the structural parameters in the model are not all identifiable. To see this, one can match the structural parameters with their reduced-form counterparts: $\tilde{\Sigma}$ with Σ and $\tilde{\phi}(B)$ with $\phi(B)$; but λ gets matched with δ and ω . Thus, one more restriction is required in order to identify all the structural parameters. A natural, though arbitrary, identifying restriction is $\sigma_{12} = 0$, so that the innovations driving the shocks to the production function, z_{1t} , are independent of the innovations driving the shocks to capital accumulation, z_{2t} . From (20), it follows that $1 + \omega = (\tilde{\sigma}_{12}/\tilde{\sigma}_{22}) - (\sigma_{12}/\sigma_{11})$ so that, if $\sigma_{12} = 0$, ω can be calculated as $1 + \omega = \tilde{\sigma}_{12}/\tilde{\sigma}_{22}$. Once ω is determined, the definition of λ in (21) implies that $\delta = \lambda/(1 + \omega)$. Given ω and δ , one can calculate $\Omega(0)$ and then use (20) to obtain $\phi(B)$ and Σ .

The model can also be estimated using the alternative-system real-wage/hours expressed in equations (14) and (16). This pair of equations can be written as

$$(22) \quad \mathbf{s}_t = \Psi(B)\mathbf{z}_t$$

where $\mathbf{s}_t = [\Delta w_t - \mu, l_t - \ln A_t]'$, and

$$(23) \quad \Psi(B) = \begin{bmatrix} \omega + [1 - \delta(1 + \omega)]B & 1 \\ 1 & 0 \end{bmatrix}$$

where, again, the factor $1/(\alpha + \omega)$ has been absorbed into z_{1t} . Just as before,

⁶Strictly speaking, we have to assume that $\tilde{\phi}(B)$ does not factor into $\phi^*(B)\tilde{\Omega}(B)$. If it did, we would have $\tilde{\phi}(B)\tilde{\Omega}(B)^{-1} = \phi^*(B)$, where $\phi^*(B)$ could be any VAR operator, so that no restrictions on the reduced form would be implied.

$\det[\Psi(B)] = -1$ so that s_t has the constrained VAR($p+1$) representation:

$$(24) \quad \bar{\phi}(B)\bar{\Psi}(B)^{-1}s_t = \bar{a}_t$$

where

$$(25) \quad \bar{\phi}(B) = \Psi(0)\phi(B)\Psi(0)^{-1}$$

$$\bar{\Psi}(B) = \Psi(B)\Psi(0)^{-1}$$

$$\bar{a}_t \sim \text{i.i.d. } \mathcal{N}[0, \bar{\Sigma}]$$

$$\bar{\Sigma} = \Psi(0)\Sigma\Psi(0)$$

and

$$\bar{\Psi}(B)^{-1} = \begin{bmatrix} 1 & -\gamma B \\ 0 & 1 \end{bmatrix}$$

$$\gamma = 1 - \delta(1 + \omega).$$

Since the reduced forms for both the output/hours and the real-wage/hours models are of the same form, the same computer program can be used to estimate both models. However, the calculation of the structural parameters, assuming again that $\sigma_{12} = 0$, is now $\omega = \bar{\sigma}_{12}/\bar{\sigma}_{22}$, and $\delta = (1 - \gamma)/(1 + \omega)$. Once ω and δ are calculated, one can obtain $\Psi(0)$ from (23) and then use (25) to calculate $\phi(B)$ and Σ .

The theory does not provide guidance regarding the length of the basic time unit and, hence, about the frequency of the observations to use in the estimation. We opted for using annual or quarterly data as described below for the two alternative ways of interpreting the model.

The results for the output/hours system are reported first, and then the real-wage/hours system is briefly addressed. The use of wage data as a proxy for the theoretical spot return on labor effort is admittedly problematic. In spite of this, we reestimated the model with real wages as a way to provide additional evidence about the model.

The estimation was carried out with U.S. data. The variable Y_t is measured by real GNP, and L_t is total employment times

average weekly hours. Since the model is based on a representative agent and constant population, both output and total hours were divided by the working-age population (between 16 and 64 years of age). The labor data are from the Current Population Survey, which is a survey of households. The real wage is measured by nominal average hourly earnings in total private nonagricultural establishments divided by the GNP deflator. The quarterly data are seasonally adjusted.⁷

The annual results for the output-growth/hours system in equation (19) are reported first. The sample period is 1954–1987. The estimation requires the specification of the order of the polynomial $\bar{\phi}(B)$, which is equal to the order of $\phi(B)$. We picked $p = 2$ given that, on the basis of the Box-Jenkins identification procedure, the univariate annual hours series that is theoretically proportional to z_{1t} appears to be described very well as an AR(2) process. Hence, to allow for the possibility that z_{1t} and z_{2t} do not interact, at least a second order for $\phi(B)$ is required.

The maximum-likelihood estimates are presented in Table 1. The restrictions implied by the model on the reduced form, tested against the alternative of an unrestricted VAR of order 3, produce a likelihood-ratio test statistic of 0.80. The 5-percent critical value is $\chi^2_{[3]} = 7.8$, so that the restrictions can be easily accepted. In similar tests performed with the three other data sets, it was likewise not possible to reject the restrictions. These results indicate that the restrictions are consistent with the data, which of course does not necessarily mean that the model is true. We prefer to see the model as a tool with which to organize and theoretically interpret the data. Another way of “testing” the model is then

⁷The only series unavailable in seasonally adjusted form is that for the average hours. It was adjusted by regressing the log of hours on four seasonal dummies. Adding to the residuals the average coefficient on the four dummies and taking the exponent produced the adjusted series.

TABLE 1—GNP/HOURS DATA SET, ANNUAL OBSERVATIONS: 1954–1987

Reduced-Form Estimates:

$$\hat{\lambda} = 0.84 \\ (0.25)$$

$$\hat{\phi}(B) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.37 & 0.28 \\ (0.28) & (0.45) \end{bmatrix} B - \begin{bmatrix} 0.18 & -0.37 \\ (0.28) & (0.33) \end{bmatrix} B^2$$

$$\hat{\Sigma} = \begin{bmatrix} 0.00042 & \\ 0.00025 & 0.00019 \end{bmatrix}$$

Structural-Parameter Estimates:

$$\hat{\omega} = 0.28 \quad \hat{\delta} = 0.66 \\ (0.12) \quad (0.26)$$

$$\hat{\phi}(B) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1.17 & 0.24 \\ -0.74 & 0.06 \end{bmatrix} B - \begin{bmatrix} -0.66 & -0.28 \\ 0.70 & 0.54 \end{bmatrix} B^2$$

$$\hat{\Sigma} = \begin{bmatrix} 0.00019 & \\ 0.0 & 0.00011 \end{bmatrix}$$

Likelihood-ratio test statistic = 0.8

Note: Standard errors are in parentheses.

to judge whether the interpretation is sensible.

Using the identifying assumption $\sigma_{12} = 0$, the structural parameter values are $\omega = 0.28$ and $\delta = 0.66$. The estimate of ω indicates a high elasticity of labor supply with respect to the real wage. This elasticity corresponds to $1/\omega \approx 3.6$. In micro-data studies (see e.g., Thomas MaCurdy, 1981; Orley Ashenfelter, 1984), the estimates of a similar elasticity are much smaller. However, most of the variation in total hours comes from movements in the number of workers rather than in hours per worker. Since the macro-data used here capture the total variation in hours, one would expect a higher estimate for that elasticity. The estimate of δ does not have, to our knowledge, comparable values in the literature. Recall that the capital-accumulation equation is $K_{t+1} = A_1 K_t^\delta I_t^{1-\delta}$, and hence $1-\delta$ represents the elasticity of the next-period capital stock with respect to current investment. However, rough comparison can be made with

the corresponding elasticity in the standard linear capital-evolution equation. In the latter case, the elasticity equals the ratio of gross investment to the usual measure of the capital stock. This ratio is in the order of magnitude of 0.1, much smaller than the $1-\delta$ value of 0.37 obtained here. Hence, the present estimate indicates higher relative importance of current investment for future productive capacity than under the usual capital accounting.

The estimation with quarterly data requires the determination of the order of $\hat{\phi}(B)$ for this case. Given the thinner choice of p to be made now, we used here the criterion suggested by E. J. Hannan (1980) for the determination of the order of an ARMA process.⁸ Surprisingly, the best

⁸The procedure is to find the integer $k = p + 1$ that minimizes $\ln[\det(\hat{\Sigma}_k)] + 8k \ln[\ln(N)]/N$, where N is the number of observations, and $\hat{\Sigma}_k$ is the estimated variance-covariance matrix of the innovations from the unconstrained vector autoregression of order k .

TABLE 2—GNP/HOURS DATA SET, QUARTERLY OBSERVATIONS:
1954:1–1987:4*Reduced-Form Estimates:*

$$\hat{\lambda} = 0.80 \\ (0.32)$$

$$\hat{\phi}(B) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.29 & 0.80 \\ (0.09) & (0.32) \end{bmatrix} B - \begin{bmatrix} 0.15 & -0.42 \\ (0.08) & (0.15) \\ 0.28 & -0.26 \\ (0.10) & (0.25) \end{bmatrix} B^2$$

$$\hat{\Sigma} = \begin{bmatrix} 0.00009 & \\ 0.00004 & 0.00012 \end{bmatrix}$$

Structural-Parameter Estimates:

$$\hat{\omega} = -0.68 \quad \hat{\delta} = 2.50 \\ (0.14) \quad (0.47)$$

$$\hat{\phi}(B) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.72 & 0.53 \\ 0.66 & 0.12 \end{bmatrix} B - \begin{bmatrix} -0.17 & 0.28 \\ -0.31 & 0.06 \end{bmatrix} B^2$$

$$\hat{\Sigma} = \begin{bmatrix} 0.0004 & \\ 0.0 & 0.0001 \end{bmatrix}$$

Likelihood-ratio test statistic = 2.6

Note: Standard errors are in parentheses.

choice turns out to be 3, implying again $p = 2$.

The results from the quarterly estimation are reported in Table 2. The likelihood-ratio test statistic is 2.6 (compared with the 5-percent critical value of 7.8). However, a problem for the model with these results is that the estimates of ω and δ are -0.68 and 2.5 , which sharply contradict the theory. The parameter ω is theoretically constrained to be positive, and δ should be between 0 and 1. The assumption responsible for those estimates is $\sigma_{12} = 0$. This assumption implies that $\tilde{\sigma}_{11} = (1 + \omega)\sigma_{11} + \sigma_{22}$ and $\tilde{\sigma}_{22} = \sigma_{11}$. Note, however, that $\tilde{\sigma}_{11} < \tilde{\sigma}_{22}$ in Table 2. Hence, if $\sigma_{12} = 0$, ω has to be negative. Then, given $\hat{\lambda}$, the estimate $\hat{\delta}$ results in a value greater than 1.

The empirical results can be reconciled with theoretically acceptable values for ω and δ if σ_{12} is negative rather than zero. This would allow for ω to be positive, which would result in a value for δ less than $0.80(\hat{\lambda})$. We return to this point below.

TABLE 3—REAL-WAGE/HOURS DATA SET

Annual Observations, 1954–1987:

$$\hat{\omega} = 0.15 \quad \hat{\delta} = 1.31 \\ (0.12) \quad (0.59)$$

Likelihood-ratio test statistic = 7.3

Quarterly Observations, 1964:1–1987:4:

$$\hat{\omega} = 0.02 \quad \hat{\delta} = 0.97 \\ (0.05) \quad (0.19)$$

Likelihood-ratio test statistic = 2.3

Note: Standard errors are in parentheses.

Results from the estimation of the real-wage/hours system in equation (24) are reported in Table 3. The quarterly series on wages is available only from 1964. Again, the restrictions cannot be rejected at the 5-percent level, although with annual data the statistic is 7.3, close to the critical value of 7.8. The table also reports the estimated structural parameters under $\sigma_{12} = 0$. With

TABLE 4—ANNUAL OUTPUT AND HOURS: SECOND MOMENTS

Lag	Variable	Model		United States data 1954–1987	
		GNP	Hours	GNP	Hours
0	GNP	1.00	0.84	1.00	0.89
	Hours	0.84	1.00	0.89	1.00
1	GNP	0.44	0.50	0.56	0.63
	Hours	0.31	0.53	0.34	0.49
2	GNP	–0.12	–0.15	0.04	0.01
	Hours	–0.20	–0.13	–0.17	–0.20
Standard deviation:		0.021	0.019	0.023	0.019

Notes: Both the actual and the model data are Hodrick-Prescott filtered. The model's results are average moments across 200 artificial samples of 34 observations.

quarterly data, the estimates are $\hat{\omega} = 0.02$ and $\hat{\delta} = 0.97$, which are (barely) within the theoretically acceptable region. The corresponding annual estimates are $\hat{\omega} = 0.15$ and $\hat{\delta} = 1.31$. Here $0 < \hat{\delta} < 1$ is violated. It is interesting to note that, if σ_{12} is negative rather than zero, the values for ω will be larger and those for δ will be smaller. This would shift the quarterly estimates further into the theoretical region and move the annual estimates into that region.

The results above imply that, for the present model to be a reasonable description of all the four data sets used, the innovations a_{1t} and a_{2t} should be negatively correlated, rather than independent.⁹ However, this would imply that the annual output/hours estimate $\hat{\omega} = 0.28$, which seems low but which is still theoretically reasonable under $\sigma_{12} = 0$, should be taken only as a lower bound. Hence, the elasticity of labor supply to the real wage emerging from that data set should be lower than the corresponding value of 3.6. Also, $\hat{\delta} = 0.66$ would be considered in this case only as an upper bound.

An additional procedure to evaluate the empirical performance of the model is to

compare the second moments from the model to the corresponding ones from the data. This procedure, common in the real-business-cycle literature, has the advantage of detecting the dimensions along which the model succeeds or fails. In the present context, however, one can point out an aspect in which the model fails, even prior to the implementation of that procedure. The model predicts that output, consumption, and investment are perfectly correlated, an implication that is obviously rejected by the data. However, it is fair to say that this weakness is due to the specific set of functional forms used and not to the general framework of endogenous growth.

Regarding output and hours, the theoretical comovement of these two variables follows from the estimates of a weakly restricted bivariate VAR of output growth and hours. Hence, the model's calculated second moments involving first-differenced output and hours should be very similar to those from the data. A more meaningful procedure reported here is to filter output and hours using the Hodrick-Prescott filter. In this case, output generated from the model contains a unit root, although no such restriction is imposed on the data. However, the model's output/hours cross-correlations are still fairly close to those from the data. Table 4 reports results for the annual model.¹⁰

⁹This negative correlation could be generated with a different framework in which z_{1t} reflects endogenous capital utilization as in Greenwood et al. (1988). In such a framework, as K_t increases due to an innovation to z_{2t} , the rate of capital accumulation declines (a negative innovation to z_{1t}), because the marginal cost of utilization depends positively on the stock of capital. The total flow of capital services, however, still increases.

¹⁰Another variable in the model is the capital stock. However, given its postulated evolution, there is no empirical counterpart to this variable.

V. Stochastic Detrending and Long-Term Impulse Responses

The present model provides a new angle for analyzing the question of how persistent output fluctuations are, which can also be posed in terms of how the movements in actual output relate to those in its long-run trend. The existing evidence on this issue varies. Nelson and Plosser (1982) and Campbell and Mankiw (1987) found little or no evidence of temporary output fluctuations, while results in Watson (1986) and Cochrane (1988) do indicate that a large part of output movements are reversed later on.

The theoretical structure of this model, within the bivariate setup, can be used to obtain another insight into the decomposition of output into permanent and transitory movements. To see this, it is convenient to rewrite equations as follows:

$$(13) \quad \Delta y_t = \mu + \frac{1 + \omega}{\alpha + \omega} (1 - \delta B) z_{1t} + z_{2t}$$

$$(14) \quad l_t = \ln(A_t) + \frac{1}{\alpha + \omega} z_{1t}.$$

Equation (13) says that output movements are generated by two types of shocks: z_{1t} and z_{2t} . Abstracting from the exogenous stochastic properties of these shocks, the model implies that an output movement generated by z_{1t} is partly temporary, given that $0 < \delta < 1$, and that when it is generated by z_{2t} it is permanent. The theoretical interpretation of these different impacts is that z_{1t} affects the market efficiency of labor, and hence it induces a fluctuation due to temporary labor effort. The reason why only a part of this fluctuation is reversed (z_{1t-1} is multiplied by δ , which is less than 1) is because capital accumulates following the shock. By contrast, z_{2t} augments permanently the productivity of labor in both market and home activities, without generating a movement in labor effort. Hence, this type of output movement is permanent.

Given the different implications of the two shocks, the bivariate setup in (13) and

(14) is particularly useful. Since hours move proportionally to z_{1t} in (14), the hours series makes it possible to identify the output movements that have a temporary component.

To measure the degree of persistence of output movements, the bivariate system was used to compute the long-run impulse responses to the structural innovations. This procedure was followed up by computing a stochastic trend in output. The reported results correspond to the annual data and estimates. It is important to note that these calculations are based on the reduced form (19), which *does not* depend on the identification of ω and δ .

The long-term impulse response of a unit size innovation corresponds to the sum of coefficients in the possibly infinite moving-average representation of the growth rates. To obtain such a representation, one can rewrite equation (19) as

$$(26) \quad \mathbf{x}_t = \Psi(B) \tilde{\mathbf{a}}_t$$

where $\Psi(B) = \tilde{\Omega}(B)\tilde{\Phi}(B)^{-1}$ and $\Psi(0) = \mathbf{I}$. Let $\psi_{ij}(B) = \sum_{k=0}^{\infty} \psi_{ij}^k B^k$ be the (i, j) th component of $\Psi(B)$. Then, for output we have

$$(27) \quad \Delta y_t = \psi_{11}(B) \tilde{a}_{1t} + \psi_{12}(B) \tilde{a}_{2t}$$

with $\tilde{a}_{1t} = (1 + \omega)a_{1t} + a_{2t}$ and $\tilde{a}_{2t} = a_{1t}$ [from equation (19)]. Hence, (27) can be expressed as

$$\Delta y_t = \left[\psi_{11}(B) + \frac{\psi_{12}(B)}{1 + \omega} \right] (1 + \omega)a_{1t} + \psi_{11}(B)a_{2t}.$$

Since $\psi_{11}(0) = 1$ and $\psi_{12}(0) = 0$ [from $\Psi(0) = \mathbf{I}$], an innovation in output of, say, 1 percent corresponds to either $(1 + \omega)a_{1t}$ or a_{2t} of the same magnitude (or a combination of both). The calculation of the long-term effect of each one of these shocks requires the values of $\psi_{1j}(1) = \sum_{k=0}^{\infty} \psi_{1j}^k$, $j = 1, 2$. These values, which take into account the total impact of a current innovation on growth rates over the entire infinite future, are calculated using the estimated parameters as $\Psi(1) = \tilde{\Omega}(1)\tilde{\Phi}(1)^{-1}$.

The long-term effect of the innovation $(1 + \omega)a_{1t}$ turns out to be 0.53. For comparison, with a process following a deterministic trend plus stationary fluctuations, the long-term response is zero. In the random-walk case, the corresponding value is 1, since the long-run level of the variable shifts with the actual level. The value of 0.53 indicates an intermediate case in which about half of the output innovation is “undone” in the future. Hence, production-function shocks generate output movements that are largely temporary.

The situation is very different with respect to the capital innovation a_{2t} . For this shock, the long-term response is 2.4. Hence, the effect of a given innovation on output due to this source builds up over time to more than twice the size of its initial impact.

The procedure followed to compute the detrended output series, which is closely related to the calculation of the long-term impulse responses, is based on a bivariate counterpart of that suggested by Beveridge and Nelson (1981) as discussed in King et al. (1989).

Detrending is achieved by the decomposition of the level of y_t into a permanent and a cyclical component:

$$y_t = y_t^p + y_t^c$$

where y_t^p is a random walk (plus drift) representing the stochastic trend and y_t^c is a stationary component. The decomposition is obtained by rewriting equation (27) as

$$(28) \quad y_t = \left[\psi_{11}(1) \frac{\tilde{a}_{1t}}{1-B} + \psi_{12}(1) \frac{\tilde{a}_{2t}}{1-B} \right] + \left[\frac{\psi_{11}(B) - \psi_{11}(1)}{1-B} \tilde{a}_{1t} + \frac{\psi_{12}(B) - \psi_{12}(1)}{1-B} \tilde{a}_{2t} \right].$$

The first expression in brackets is a random walk and is thus defined as the permanent component y_t^p . It is related to the long-run

impulse responses because it represents the long-run level of output (except for the deterministic growth).

The expression in the second set of brackets in (28) is defined as y_t^c . Using the fact that the residuals $\tilde{a}_{1t}$ and $\tilde{a}_{2t}$ can be written in terms of the observable variables, it is shown in Appendix B that y_t^c can be expressed as

$$(29) \quad y_t^c = \Gamma_1(B) \Delta y_t + \Gamma_2(B) l_t$$

where $\Gamma_1(B)$ and $\Gamma_2(B)$ are finite-order polynomials whose coefficients can be calculated from the estimated parameters. Since Δy_t and l_t are stationary, equation (29) implies that y_t^c is also stationary. Once y_t^c is calculated from (29), the stochastic trend can be obtained as $y_t^p = y_t - y_t^c$.

This method of calculating the permanent and transitory components differs from the procedure suggested by Beveridge and Nelson. Their procedure is to calculate y_t^p by recursively producing forecasts of the future levels of the time-series, which approach the trend value as the forecast period increases. Here, given the vector autoregressive form, we can obtain and calculate closed-form expressions for the trend and the transitory component.

Figure 1 depicts the stochastic trend along with the actual values of output. The resulting values for y_t^c and the cyclical movement in total hours, which is l_t itself minus a constant, are plotted in Figures 2 and 3. The cycles in hours exhibit troughs in the years 1958, 1961, 1964, 1971, 1975, and 1982, most of them matching the conventional chronology of business cycles. These fluctuations tend also to be reflected, to some extent, in the cycles of output depicted in Figure 2.

To illustrate the model's interpretation of the detrending and impulse-response calculations it is of particular interest to consider the markedly different behavior of detrended output and hours in the 1974–1975 episode. Figure 3 shows that hours turn downward slightly in 1974, and then they decline sharply in 1975. In contrast, detrended output surprisingly turns upward in

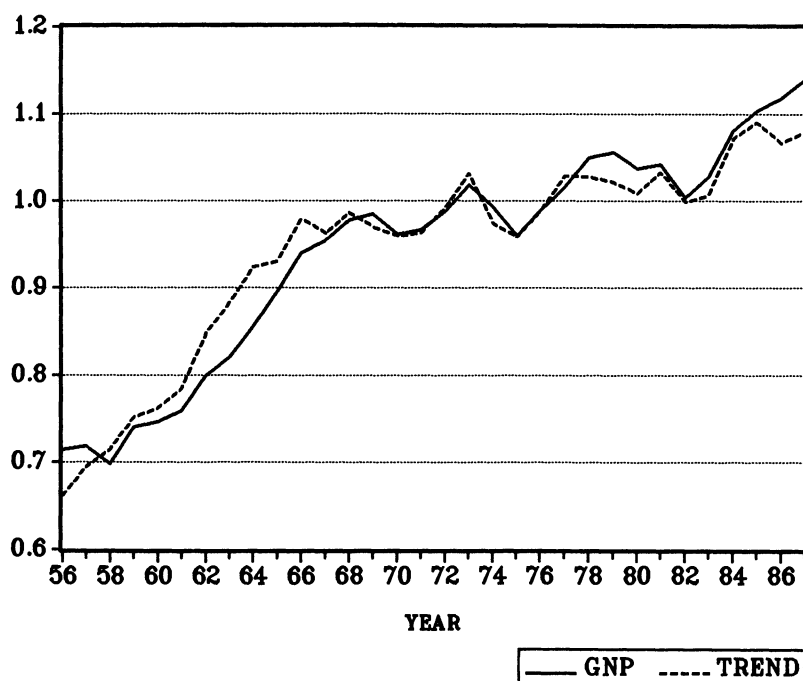


FIGURE 1. GNP PER CAPITA AND TREND

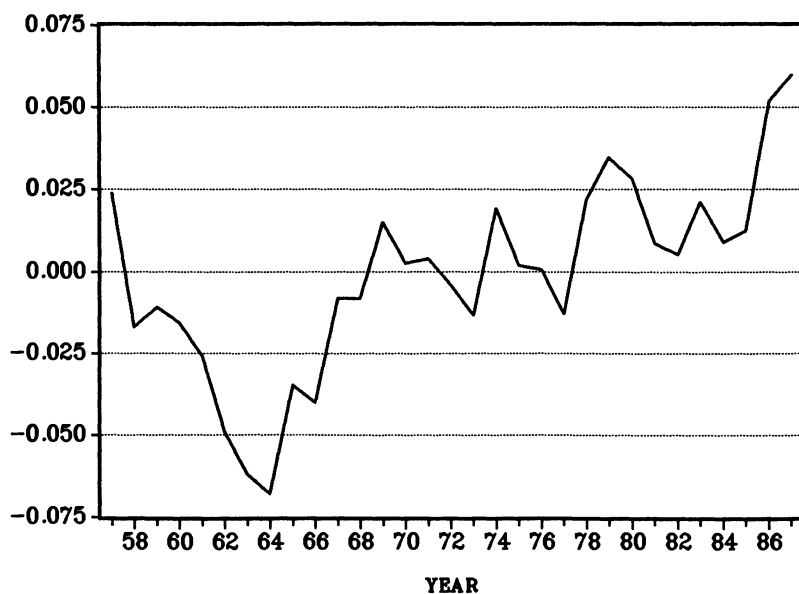


FIGURE 2. DETRENDED PER CAPITA GNP

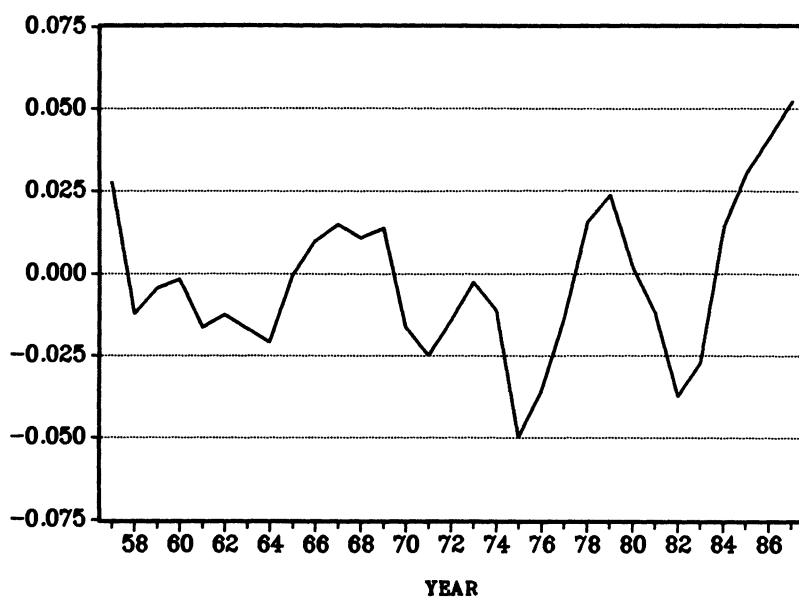


FIGURE 3. HOURS PER CAPITA (DEVIATIONS FROM AVERAGE)

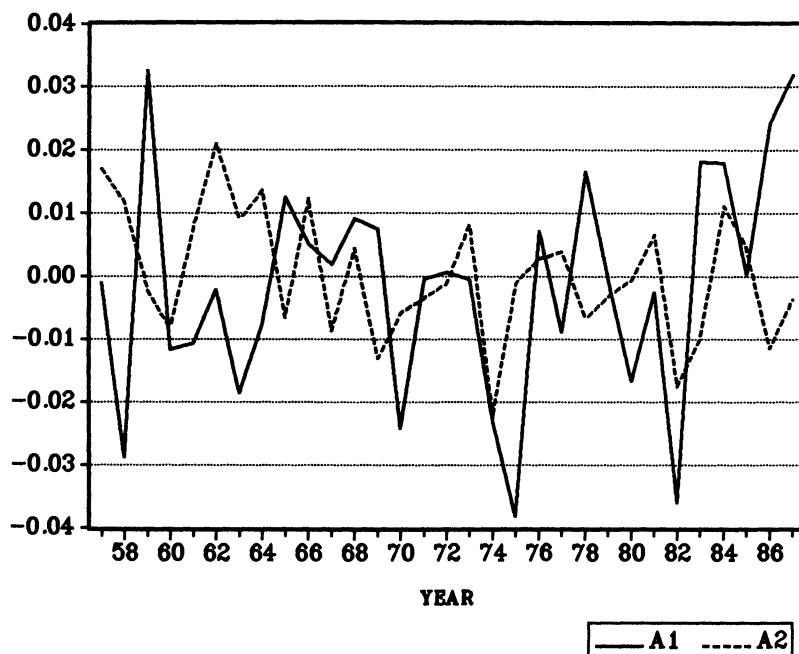


FIGURE 4. INNOVATIONS

1974 and only declines in 1975, but not nearly as sharply as do hours (Fig. 2).

The source of the difference lies in the sharp negative innovation to capital accumulation in 1974. Note in Figure 4 that the value of a_{2t} is below -2 percent, the lowest value in the sample. This observation corresponds to the first oil shock which can be interpreted as rendering part of the capital stock obsolete at the higher oil prices. Given the long-run impulse response of 2.4 to capital-accumulation innovations (see above) the long-run value of output (or trend) declines by about 5.5 percent in 1974 (this is the calculated permanent loss of annual output due to that particular negative shock). Now, for the measurement of the temporary component, what matters is that the trend declines by more than actual output (see Fig. 1), producing the computed increase in detrended output in 1974. Then, the observation of 1975 is dominated by the productivity innovation of -3.8 percent, also the lowest in the sample (see Fig. 4), which generates a decline in detrended output. Recall that for this type of disturbance the long-run effect is 0.52, so that the trend responds by less than does actual output.

Another interesting episode is 1986–1987. In both years, output growth and hours are high. The model translates these movements into large values (2.4 percent and 3.2 percent) for the innovation a_{1t} , which has temporary effects.

Informally, the bivariate output/hours detrending procedure amounts to the following. In a given period, if hours move in the same direction and by a similar percentage as output, the model takes this as an underlying a_{1t} innovation. Hence, the output change is predicted to be largely transitory. If hours move significantly less than output, as in the period 1962–1964 and 1974, the model interprets this as an underlying a_{2t} innovation which has strong permanent effects. In these cases, the trend shifts more than actual output.

VI. Concluding Remarks

This paper studied the determination of output, the real wage, and hours of work in a real-business-cycle model with endoge-

nous growth. As in Arrow (1962) and Romer (1986), a characteristic of the model is a positive externality associated with investment activity. Investment is seen as producing the accumulation of knowledge, which in turn increases productivity at the social level. This form of technological progress is thought of as also affecting the productivity in home activities, which in turn produce utility.

The model is estimated in two forms: output/hours and real-wage/hours. Annual and quarterly data were used alternatively. The restrictions imposed by the model were not rejected in any of the data sets. These results are not taken as evidence that the model is appropriate, but that the restrictions are weak. We prefer to think of the model as a tool with which to interpret the data. Whether the interpretation is sensible or not is then a way of evaluating the model.

The way the model organizes the data can be summarized as follows. Hours of work depend only on the productivity in market activities relative to that in home activities. This feature has two main consequences. One is that the stationarity of the process of hours per capita is not due to an income effect, but to parallel long-run technological progress in market and home activities. The other consequence is that real wage movements do not have to be temporary to generate labor fluctuations. The processes of output and the real wage, however, contain unit roots because of the endogenous growth mechanism. Shocks have permanent effects both on output and the real wage through their effects on capital and knowledge.

The model provides a new framework for performing and interpreting a bivariate calculation of the persistence of output fluctuations. According to the model, output fluctuations are caused by two types of shocks. One is a disturbance to the production function, which has transitory effects since it alters the relative productivity of labor in market and home activities, and the other is a shock to capital accumulation, which has only permanent effects. Hours of work fluctuate with the first shock, and hence it is possible to identify those output movements that contain a large transitory

component. Hence, the bivariate output/hours setup is a useful framework for carrying out the decomposition of output into permanent and transitory components.

APPENDIX A

Consider a model identical to that in the text except that it has the criterion function in (5):

$$U = \ln[C_t + H_t(L - L_t^{1+\omega})].$$

Given that this model does not have a closed-form solution, as does that in the text, a different strategy should be adopted to solve for the equilibrium decision rules.

Define $\tilde{X}_t = X_t/H_t$, so that the model can be transformed into one that possesses a deterministic steady state (see e.g., Christiano and Eichenbaum, 1988; King et al., 1988). In the transformed economy, the planner's problem is to maximize

$$(A1) \quad E_t \sum_{j=0}^{\infty} \beta^j [\ln(\tilde{C}_{t+j} + L - L_{t+j}^{1+\omega}) + \ln H_{t+j}]$$

subject to

$$(A2) \quad \tilde{Y}_t = A_0 \tilde{K}_t^\alpha \tilde{L}_t^{1-\alpha} \exp(z_{1t})$$

$$(A3) \quad \tilde{K}_{t+1} = A_1 \frac{H_t}{H_{t+1}} \tilde{K}_t^\delta \tilde{I}_t^{1-\delta} \exp(z_{2t+1})$$

$$(A4) \quad \tilde{Y}_t = \tilde{C}_t + \tilde{I}_t.$$

The first-order optimality conditions are

$$(A5) \quad (1-\alpha)A_0 \tilde{K}_t^\alpha \tilde{L}_t^{-\alpha} \exp(z_{1t}) \\ = (1+\omega)L_t^\omega$$

$$(A6) \quad \frac{1}{\tilde{Y}_t - \tilde{I}_t + L - L_t^{1+\omega}} \\ = \beta E_t \frac{\alpha \tilde{Y}_{t+1} / \tilde{K}_{t+1} + \left(\frac{\delta}{1-\delta}\right) \tilde{I}_{t+1} / \tilde{K}_{t+1}}{\tilde{Y}_{t+1} - \tilde{I}_{t+1} + L - L_{t+1}^{1+\omega}} \\ \times (1-\delta) \frac{\tilde{K}_{t+1}}{\tilde{I}_t}.$$

From (A5) and the production function, it follows that

$$(A7) \quad L_t = A_\ell \tilde{K}_t^{\frac{\alpha}{\alpha+\omega}} \exp(z_{1t})^{\frac{1}{\alpha+\omega}}$$

$$(A8) \quad \tilde{Y}_t = A_y \tilde{K}_t^{\frac{\alpha(1+\omega)}{\alpha+\omega}} \exp(z_{1t})^{\frac{1+\omega}{\alpha+\omega}}$$

where A_ℓ and A_y are defined in equations (8) and (9).

Since in equilibrium $\tilde{K}_t = K_t$, or $\tilde{K}_t = 1$ for all t , we have

$$(A9) \quad \frac{H_{t+1}}{H_t} = \frac{K_{t+1}}{K_t} \\ = A_1 K_t^{\delta-1} I_t^{1-\delta} \exp(z_{2t+1}) \\ = A_1 \tilde{I}_t \exp(z_{2t+1})$$

$$(A10) \quad L_t = A_\ell \exp(z_{1t})^{\frac{1}{\alpha+\omega}}$$

[which is identical to (8') in the text], and

$$(A11) \quad \tilde{Y}_t = A_y \exp(z_{1t})^{\frac{1+\omega}{\alpha+\omega}}$$

or

$$Y_t = A_y K_t \exp(z_{1t})^{\frac{1+\omega}{\alpha+\omega}}$$

[which is identical to (9') in the text]. Substituting $\tilde{K}_t = \tilde{K}_{t+1} = \tilde{K}_{t+2} = 1$, and (A9), (A10), and (A11) in the Euler equation (A6) yields

$$(A12) \quad \frac{\left(\frac{1}{1-\delta}\right) \tilde{I}_t}{\left(\frac{A_y - A_\ell^{1+\omega}}{A_y}\right) \tilde{Y}_t - \tilde{I}_t + L} \\ = \beta E_t \frac{\alpha \tilde{Y}_{t+1} + \left(\frac{\delta}{1-\delta}\right) \tilde{I}_{t+1}}{\left(\frac{A_y - A_\ell^{1+\omega}}{A_y}\right) \tilde{Y}_{t+1} - \tilde{I}_{t+1} + L}$$

where $A_y - A_\ell^{1+\omega} > 0$ [from the definitions of A_y and A_ℓ in (8) and (9)].

If $L = 0$, the model is identical to that in the text, where the solution in (10) was that investment is proportional to output: $\tilde{I}_t = b\tilde{Y}_t$. Postulating that solution here, it can be easily checked that (A12) holds, implying that $b = \alpha\beta(1-\delta)/(1-\delta\beta)$, as in (10') in the text.

Consider now the deterministic balanced-growth case, where $z_{1t} = 0$. Equation (A12) implies that

$$\left(\frac{1}{1-\delta}\right)\tilde{I} = \beta\left[\alpha\tilde{Y} + \left(\frac{\delta}{1-\delta}\right)\tilde{I}\right]$$

which again reduces to

$$\tilde{I} = \frac{\alpha\beta(1-\delta)}{1-\delta\beta}\tilde{Y}$$

or

$$I_t = \frac{\alpha\beta(1-\delta)}{1-\delta\beta}Y_t.$$

The solution for the original variables Y_t and K_t can now be calculated. From (A11), it follows that along the deterministic balanced-growth path

$$Y_t = A_y K_t$$

$$I_t = bA_y K_t$$

and, from the capital-evolution equation,

$$K_{t+1} = A_k K_t$$

where A_k is defined in equation (10). These paths correspond to those in the text [equations (9'), (10'), and (12')] with the shocks set to zero. Hence, the use of the criterion function (4) instead of (5) does not alter the deterministic-growth version of the model.

The assessment of the stochastic behavior requires the numerical simulation of the investment decision using (A12). In this equation, the current $\tilde{I}_t$ decision depends on the state variable z_{1t} [through $\tilde{Y}_t$ in (A11)] and on other variables only through

their predictive power over z_{1t+1} . Hence, the decision about $\tilde{I}_t$ depends on z_{1t} and on the process driving this shock.

The $\tilde{I}_t$ choice was simulated under two extreme cases regarding the z_{1t} process. One is when z_{1t} is independently and identically distributed, and the other is when z_{1t} is a random walk. The model in the text requires stationarity for z_{1t} , so that the random walk should be interpreted only as a limiting case. To perform the simulations, the following values were chosen for the parameters: $A_0 = 1$, $\alpha = 0.3$, and the annual data estimates $\hat{\delta} = 0.66$ and $\hat{\omega} = 0.28$. This implies that the mean of $\tilde{Y}$ (A_y) is 0.49. The standard deviation of $\tilde{Y}$ was chosen as follows. The standard deviation of $\ln L_t$ is $[1/(\alpha + \omega)]\sigma_{11}^{1/2}$, which is 0.025 in annual data. Correspondingly, the standard deviation of $\ln \tilde{Y}_t$ is $(1 + \omega)0.025 = 0.0325$. The value of 0.34 was picked for L . The average of $L_t^{1+\omega}$ is $A_\ell^{1+\omega} = 0.26$. $L^{1+\omega}$ has the interpretation of total hours of work and rest.

Case (1): z_{1t} Independently and Identically Distributed.—A grid of 100 equally spaced points for $\tilde{Y}$ with equal probabilities was chosen so that the standard deviation of $\tilde{Y}$ is 3.25 percent. An initial 100-vector guess (zeros) for $\tilde{\mathbf{I}}$ was chosen and used on the right-hand side (RHS) of (A12). With the resulting value on the RHS, another choice vector $\tilde{\mathbf{I}}$ was calculated from (A12) on the left-hand side. This $\tilde{\mathbf{I}}$ choice was used in a second iteration, obtaining a new $\tilde{\mathbf{I}}$ choice, and so on, until one-by-one convergence of the elements in $\tilde{\mathbf{I}}$.

The resulting average $\tilde{I}/\tilde{Y}$ was 0.26 with a standard deviation of 1.92 percent. When $L = 0$ (the case in the text) the ratio $\tilde{I}/\tilde{Y}$ is 0.26 exactly.

Case (2): z_{1t} Random Walk.—The same range of $\tilde{Y}$ was used, with equal $\frac{1}{3}$ probabilities of moving lower, moving higher, or staying in the same point. At the endpoints the probability $\frac{2}{3}$ was assigned to staying, and a probability of $\frac{1}{3}$ was assigned to the adjacent points. In this case, for each point on the current grid for $\tilde{Y}$ there is an expected value on the RHS. Iterations on the $\tilde{I}$ choice were carried out using (A12) until one-to-one convergence. Different grid sizes were used here: 10, 20, 50, and 100. In all

cases, the standard deviation of the $\tilde{I}/\tilde{Y}$ ratio around 0.26 is extremely small, less than 0.01 percent in the 10-grid case and even less as the grid size increases.

We view these results as follows. The true process for z_{1t} is more complicated than the two forms considered above and is jointly evolving with z_{2t} . However, given that z_{1t} is stationary but displays significant persistence, we conclude that the error in the investment choice caused by the approximate criterion (4) is far smaller than that in the white-noise case.

APPENDIX B

In order to prove equation (29), we express equation (19) as

$$(B1) \quad \phi^*(B)\mathbf{x}_t = \tilde{\mathbf{a}}_t$$

$$\phi^*(B) = \tilde{\phi}(B)\tilde{\Omega}(B)^{-1}$$

and rewrite for convenience equation (28):

$$(B2) \quad y_t = \left[\psi_{11}(1) \frac{\tilde{a}_{1t}}{1-B} + \psi_{12}(1) \frac{\tilde{a}_{2t}}{1-B} \right] + \left[\frac{\psi_{11}(B) - \psi_{11}(1)}{1-B} \tilde{a}_{1t} + \frac{\psi_{12}(B) - \psi_{12}(1)}{1-B} \tilde{a}_{2t} \right] = y_t^p + y_t^c$$

where $\psi_{ij}(B)$ is the (i, j) th component of $\Psi(B) = \phi^*(B)^{-1}$. As shown below, ignoring the constant terms will not affect the generality of the argument.

Substituting the left-hand side of (B1) for $\tilde{a}_{1t}$ and $\tilde{a}_{2t}$ in (B2), one obtains y_t^p as a

function of $(1-B)y_t$ and l_t :

$$(B3) \quad y_t^p = \psi_{11}(1) \times \left[\frac{\phi_{11}^*(B)(1-B)y_t + \phi_{12}^*(B)l_t}{1-B} \right] + \psi_{12}(1) \times \left[\frac{\phi_{22}^*(B)l_t + \phi_{21}^*(B)(1-B)y_t}{1-B} \right] = [\psi_{11}(1)\phi_{11}^*(B) + \psi_{12}(1)\phi_{21}^*(B)]y_t + \left[\frac{\psi_{12}(1)\phi_{22}^*(B) + \psi_{11}(1)\phi_{12}^*(B)}{1-B} \right]l_t = \Gamma_1^*(B)y_t + \frac{\Gamma_2^*(B)}{(1-B)}l_t$$

where $\phi_{ij}^*(B)$ is the (i, j) th component of $\phi^*(B)$. Now, from the fact that $\Psi(1) \cdot \phi^*(1) = \mathbf{I}$, it follows that

$$(B4) \quad \Gamma_1^*(1) = 1 \quad \Gamma_2^*(1) = 0$$

so that $\Gamma_2^*(B) = \tilde{\Gamma}_2(B)(1-B)$, where $\tilde{\Gamma}_2(B)$ is a finite-order polynomial. Hence,

$$(B5) \quad y_t^p = \Gamma_1^*(B)y_t + \tilde{\Gamma}_2(B)l_t.$$

Now, using the fact that $y_t^c = y_t - y_t^p$, it follows that

$$(B6) \quad y_t^c = [1 - \Gamma_1^*(B)]y_t - \tilde{\Gamma}_2(B)l_t.$$

Since $\Gamma_1^*(1) = 1$, it follows that $1 - \Gamma_1^*(B) = \Gamma_1(B)(1-B)$, where $\Gamma_1(B)$ is a finite-order polynomial. We thus have

$$(B7) \quad y_t^c = \Gamma_1(B) \cdot (1-B)y_t + \Gamma_2(B)l_t$$

$$\Gamma_2(B) = -\tilde{\Gamma}_2(B)$$

expressing the cyclical component as a finite linear combination of present and past values of $(1-B)y_t$ and l_t . If $E[(1-B)y_t] =$

$\mu \neq 0$, then one can obtain a zero mean y_t^c by substituting $\Delta y_t - \mu = \Delta y_t$ into (B7). This proves (29).

REFERENCES

- Arrow, Kenneth J.**, "The Economic Implications of Learning by Doing," *Review of Economic Studies*, June 1962, 29, 155–72.
- Ashenfelter, Orley**, "Macroeconomic Analyses and Microeconomic Analyses of Labor Supply," *Carnegie-Rochester Conference Series in Public Policy*, Autumn 1984, 21, 117–56.
- Beveridge, Stephen and Nelson, Charles R.**, "A New Approach to Decomposition of Economic Time Series into Permanent and Transitory Components with Particular Attention to Measurement of the Business Cycle," *Journal of Monetary Economics*, March 1981, 7, 151–74.
- Campbell, John J. and Mankiw, N. Gregory**, "Are Output Fluctuations Transitory?" *Quarterly Journal of Economics*, November 1987, 102, 857–80.
- Christiano, Lawrence J. and Eichenbaum, Martin**, "Is Theory Really Ahead of Measurement? Current Real Business Cycle Theories and Aggregate Labor Market Fluctuations," mimeo, Northwestern University, August 1988.
- Cochrane, John H.**, "How Big Is the Random Walk in GNP," *Journal of Political Economy*, October 1988, 96, 893–920.
- Ghez, Gilbert R. and Becker, Gary S.**, *The Allocation of Time and Goods over the Life Cycle*, New York: National Bureau of Economic Research, 1975.
- Greenwood, Jeremy and Hercowitz, Zvi**, "The Allocation of Capital and Time over the Business Cycle," mimeo, Institute for Empirical Macroeconomics, March 1990; *Journal of Political Economy*, forthcoming.
- _____, _____ and **Huffman, Gregory**, "Investment, Capacity Utilization, and the Real Business Cycle," *American Economic Review*, June 1988, 78, 402–17.
- Hannan, E. J.**, "The Estimation of the Order of an ARMA Process," *Annals of Statistics*, September 1980, 8, 1071–81.
- Hayashi, Fumio**, "Tobin's Marginal q and Average q : A Neoclassical Interpretation," *Econometrica*, January 1982, 50, 213–24.
- Heckman, James J.**, "A Life-Cycle Model of Earnings, Learning, and Consumption," *Journal of Political Economy*, August 1976, 84, S11–S44.
- King, Robert G. and Rebelo, Sergio T.**, "Business Cycles with Endogenous Growth," mimeo, University of Rochester, July 1986.
- _____, **Plosser, Charles I. and Rebelo, Sergio T.**, "Production, Growth and Business Cycles: I. The Basic Neoclassical Model," *Journal of Monetary Economics*, March/May 1988, 21, 195–232.
- _____, _____ and _____, "Production, Growth and Business Cycles: II. New Directions," *Journal of Monetary Economics*, March/May 1988, 21, 309–41.
- _____, _____, **Stock, James H. and Watson, Mark W.**, "Stochastic Trends and Macroeconomic Fluctuations," mimeo, University of Rochester, December 1989.
- Kydland, Finn E. and Prescott, Edward**, "Time to Build and Aggregate Fluctuations," *Econometrica*, November 1982, 50, 1345–70.
- Long, John B. and Plosser, Charles I.**, "Real Business Cycles," *Journal of Political Economy*, February 1983, 91, 39–69.
- Lucas, Robert E.**, "On the Mechanics of Economic Development," *Journal of Monetary Economics*, July 1988, 22, 3–42.
- _____, and **Prescott, Edward C.**, "Investment under Uncertainty," *Econometrica*, September 1971, 39, 659–81.
- _____, and **Rapping, Leonard A.**, "Real Wages, Employment, and Inflation," *Journal of Political Economy*, September/October 1969, 77, 721–54.
- MacCurdy, Thomas E.**, "An Empirical Model of Labor Supply in a Life-Cycle Setting," *Journal of Political Economy*, December 1981, 89, 1059–85.
- McCallum, Bennett T.**, "Real Business Cycle Models," in Robert J. Barro, ed., *Modern Business Cycle Theory*, Cambridge, MA: Harvard University Press, 1989, pp. 16–50.

Nelson, Charles R. and Plosser, Charles I. "Trends and Random Walks in Macroeconomic Time Series: Some Evidence and Implications," *Journal of Monetary Economics*, September 1982, 10, 139–62.

Romer, Paul M., "Increasing Returns and

Long-Run Growth, *Journal of Political Economy*, October 1986, 94, 905–26.

Watson, Mark W., "Univariate Detrending Methods with Stochastic Trends," *Journal of Monetary Economics*, July 1986, 18, 49–76.